Linear Discriminant Analysis

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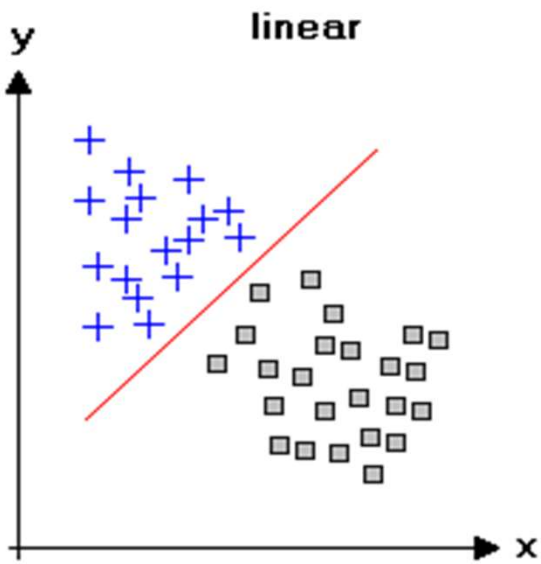
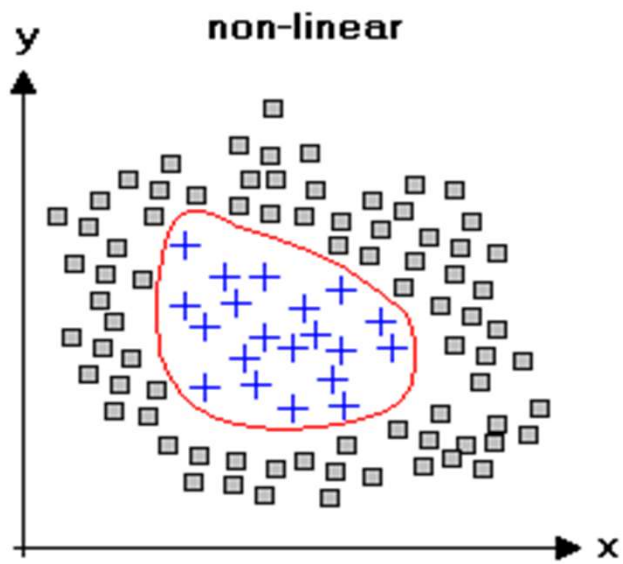
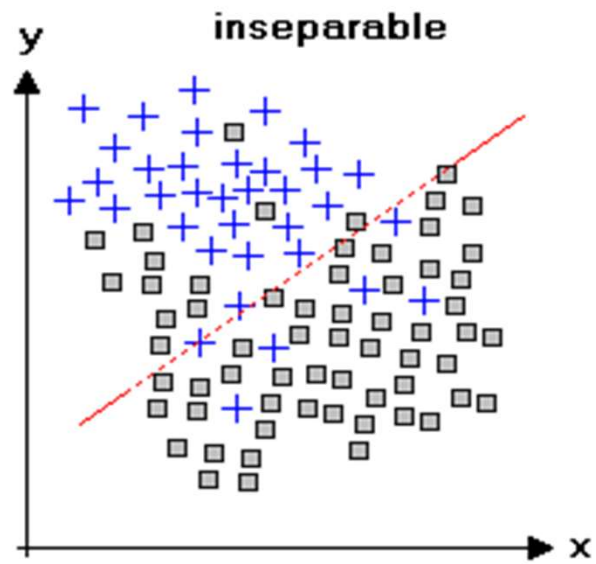
Separability in Feature Space



Linearly Separable



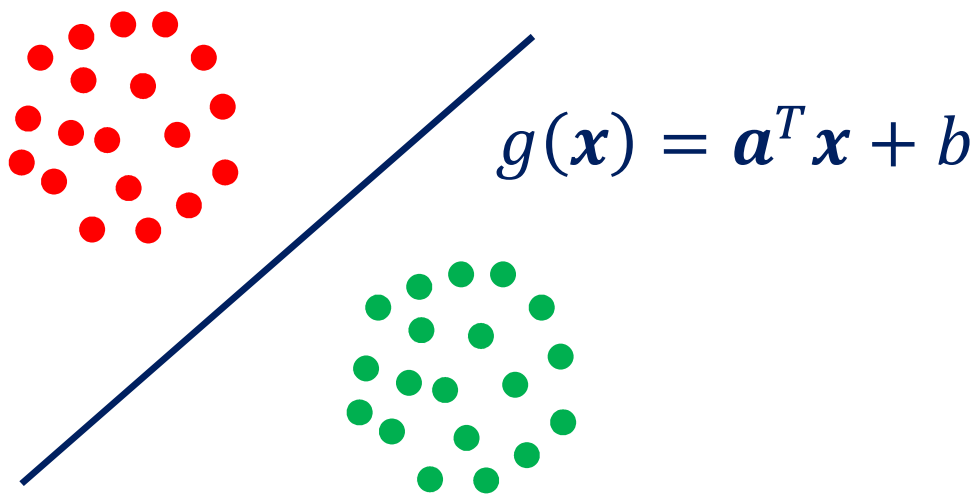
Separable by Non-Linear Boundary



Not Separable in Feature Space

Discriminant Functions: Case - I

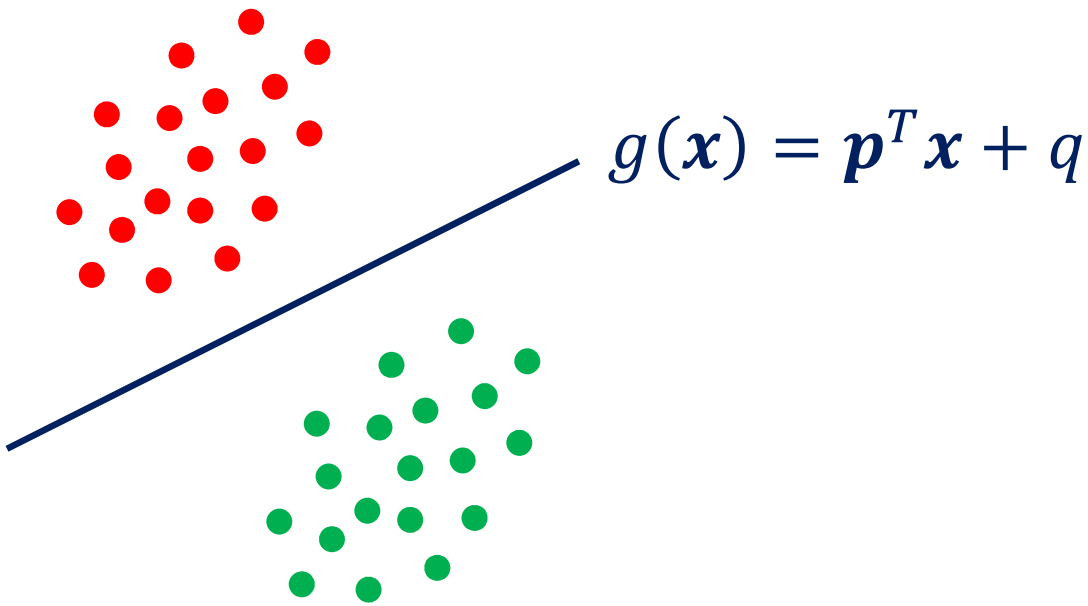
$C_0 = C_1 = \sigma^2 I$



Instances from both Classes are Normally Distributed

Discriminant Functions: Case - II

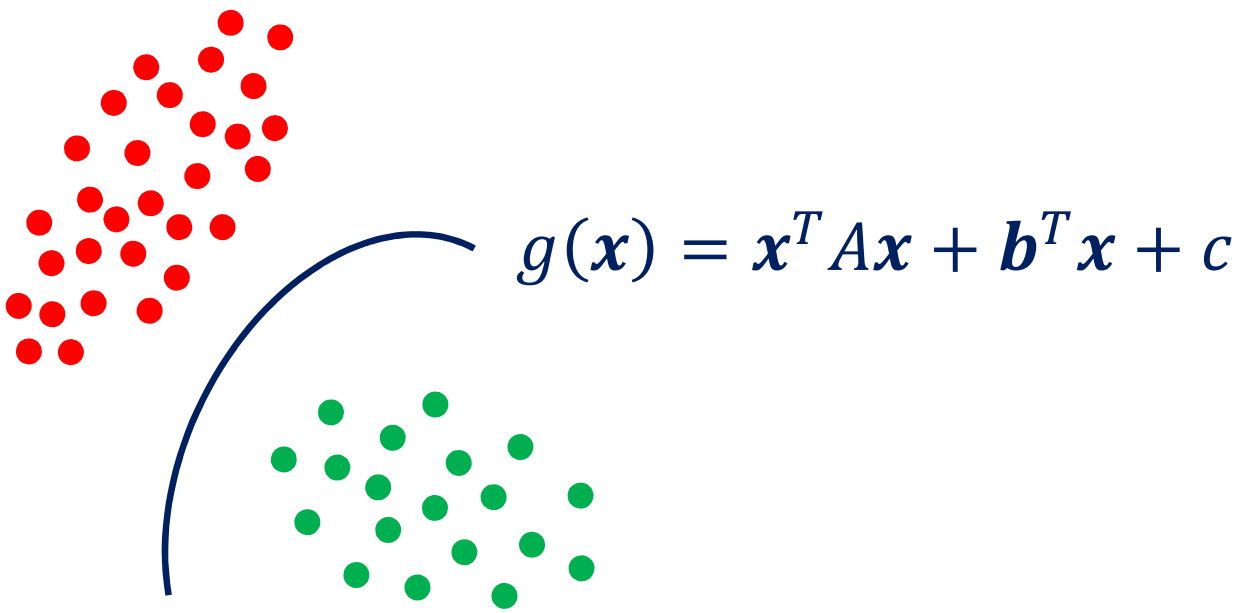
$C_0 = C_1 = C$



Instances from both Classes are Normally Distributed

Discriminant Functions: Case - II

$C_0 \neq C_1$



Instances from both Classes are Normally Distributed

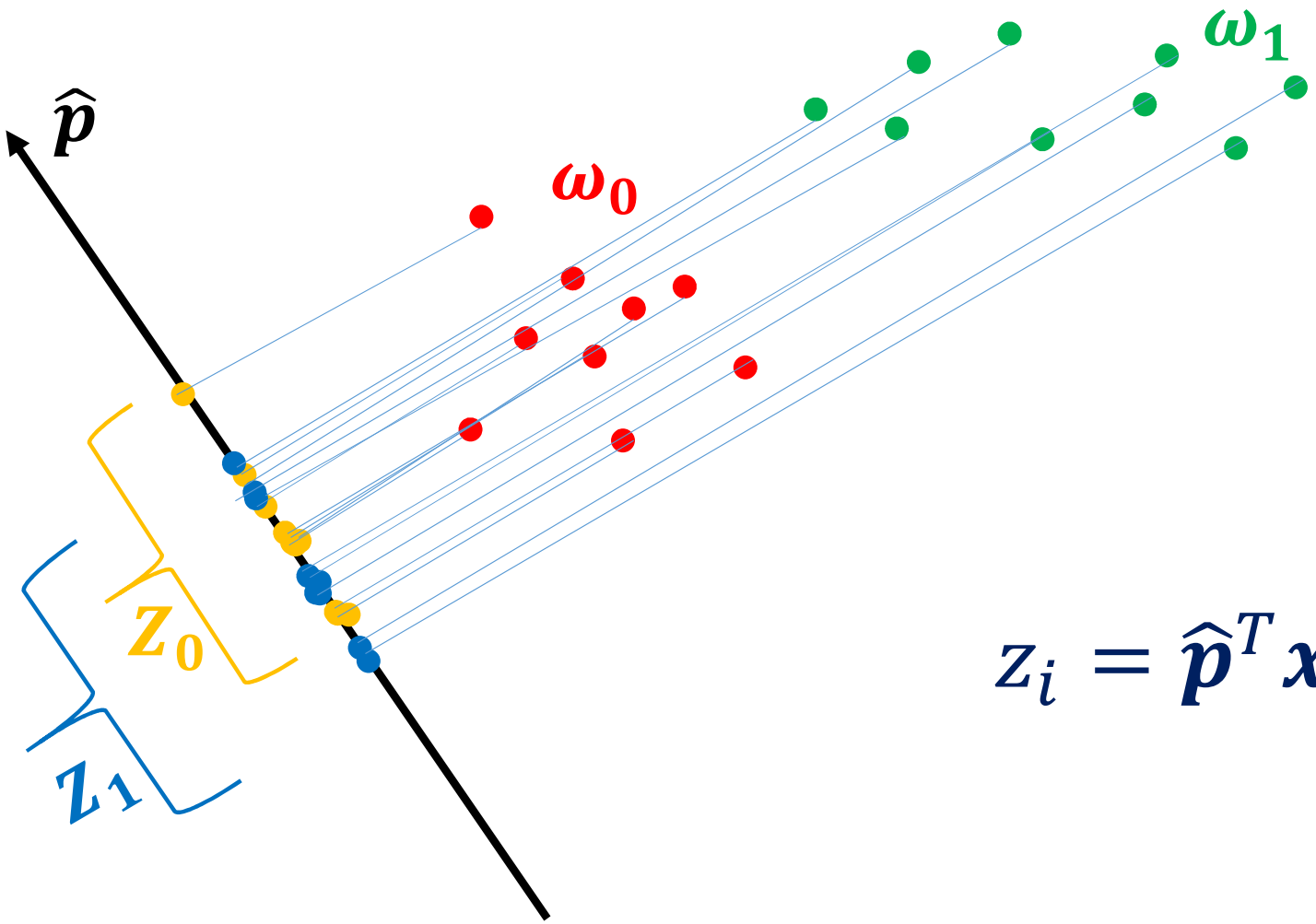
Classification: Input Data & Label

$$\boldsymbol{\omega_1} : \boldsymbol{D_1} = \{\boldsymbol{x}_j^1 : \boldsymbol{x}_j^1 \in \mathbb{R}^d ; j = 1, \dots n_1\}$$

$$\boldsymbol{\omega_0} : \boldsymbol{D_0} = \{\boldsymbol{x}_i^0 : \boldsymbol{x}_i^0 \in \mathbb{R}^d ; i = 1, \dots n_0\}$$

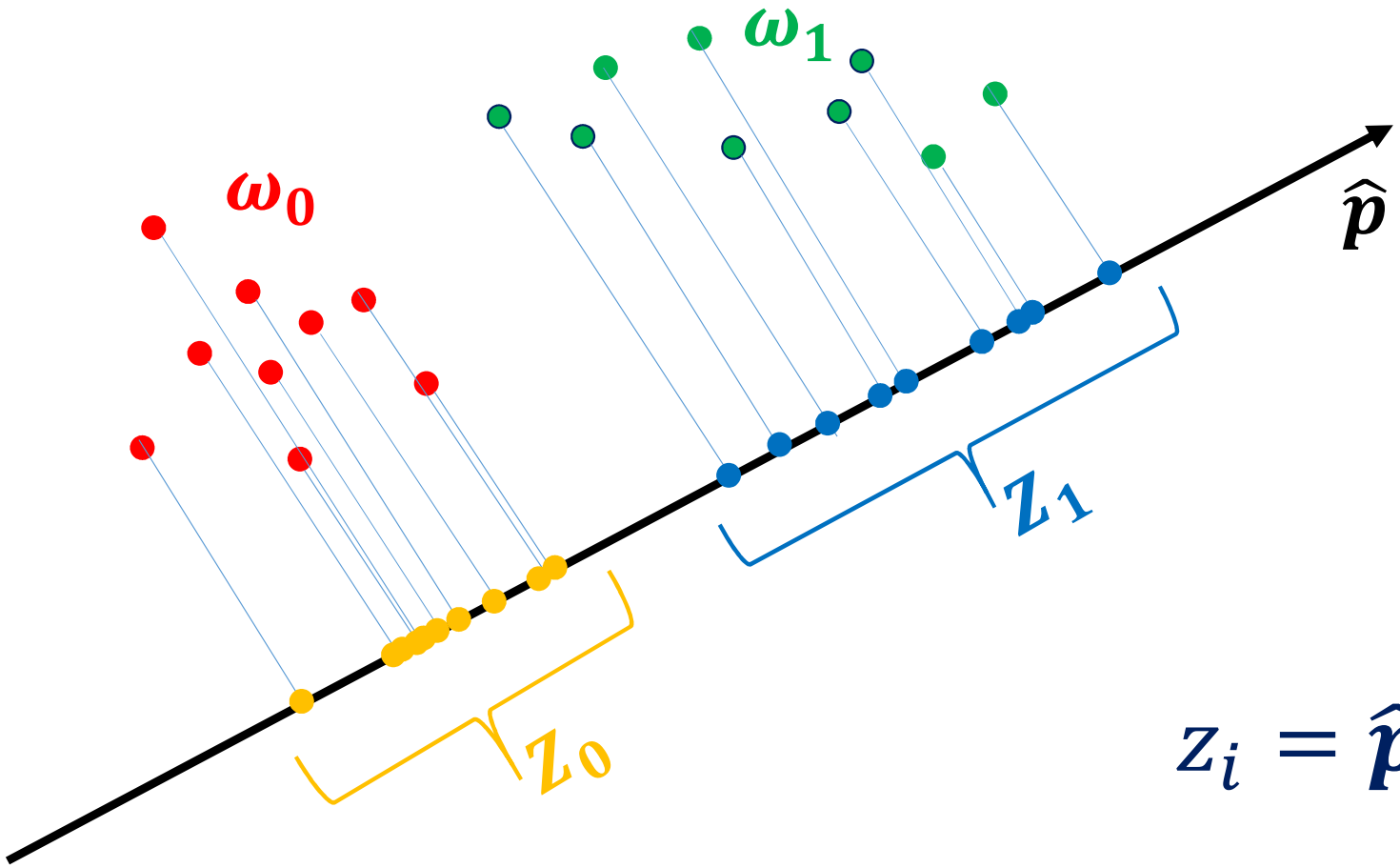
$$y(\boldsymbol{x}) = \begin{cases} 1, & \boldsymbol{x} \in \boldsymbol{\omega_1} \\ 0, & \boldsymbol{x} \in \boldsymbol{\omega_0} \end{cases}$$

Projection along a Line



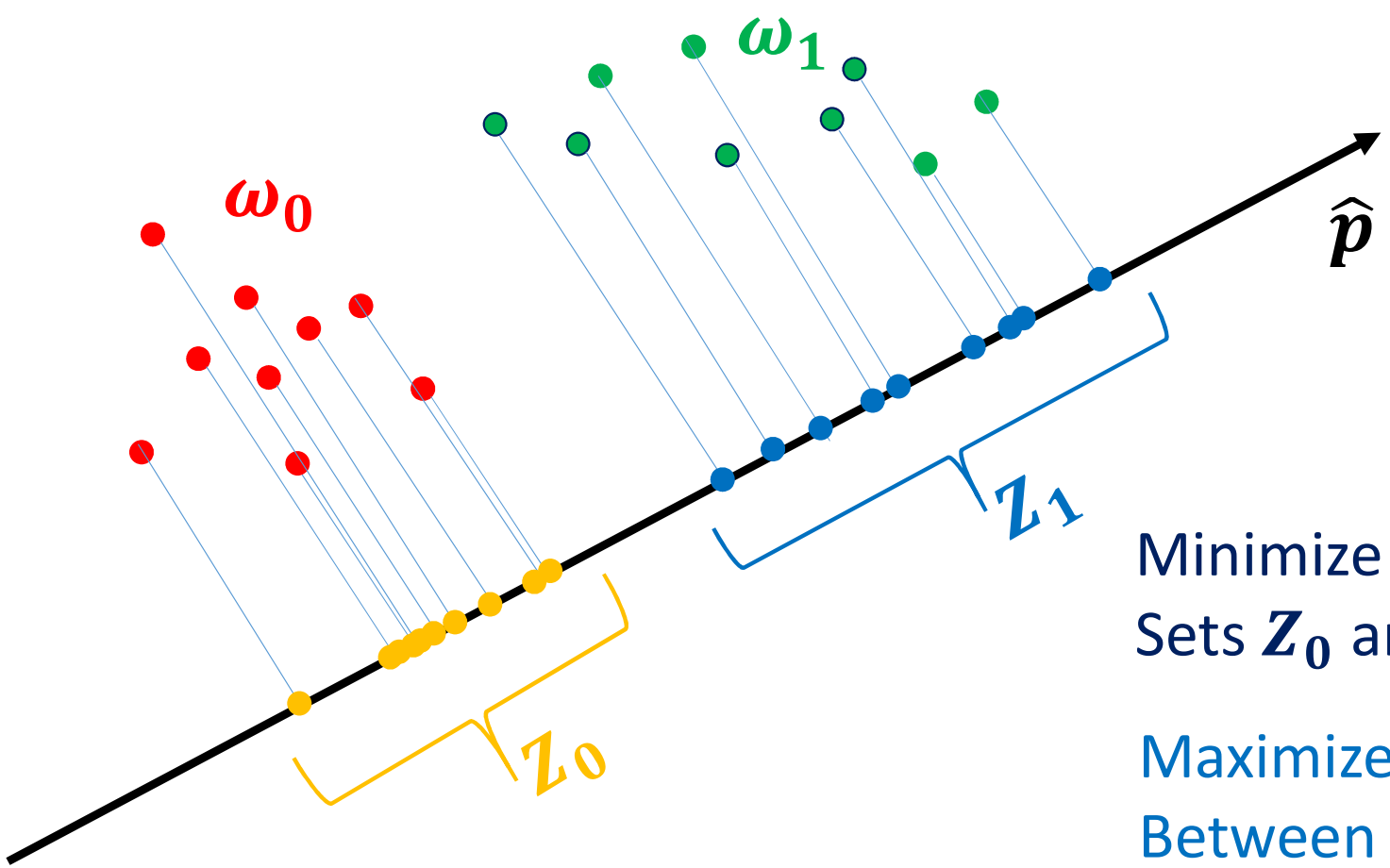
$$z_i = \hat{\boldsymbol{p}}^T \boldsymbol{x}_i$$

Projection along a Line



$$z_i = \hat{\boldsymbol{p}}^T \boldsymbol{x}_i$$

Constrained Projection along a Line



Minimize Variances of Sets Z_0 and Z_1

Maximize Distance Between Sets Z_0 and Z_1

Definitions

$$\boldsymbol{D_0} = \{\boldsymbol{x}_i^0 : \boldsymbol{x}_i^0 \in \mathbb{R}^D; i = 1, \dots n_0\}$$

$$\boldsymbol{D_1} = \{\boldsymbol{x}_j^1 : \boldsymbol{x}_j^1 \in \mathbb{R}^D; j = 1, \dots n_1\}$$

$$y(\boldsymbol{x}) = \begin{cases} 1, & \boldsymbol{x} \in \boldsymbol{\omega_1} \\ 0, & \boldsymbol{x} \in \boldsymbol{\omega_0} \end{cases}$$

$$\boldsymbol{Z_0} = \{z_i^0 : z_i^0 = \boldsymbol{p}^T \boldsymbol{x}_i^0; \boldsymbol{x}_i^0 \in \boldsymbol{D_0}\}$$

$$\boldsymbol{Z_1} = \{z_j^1 : z_j^1 = \boldsymbol{p}^T \boldsymbol{x}_j^1; \boldsymbol{x}_j^1 \in \boldsymbol{D_1}\}$$

$$\boldsymbol{p}^T \boldsymbol{p} = 1$$

$$\boldsymbol{\mu}_0 = Mean(\boldsymbol{D_0}); \boldsymbol{C}_0 = Cov(\boldsymbol{D_0})$$

$$m_0 = Mean(\boldsymbol{Z_0}); v_0 = Var(\boldsymbol{Z_0})$$

$$\boldsymbol{\mu}_1 = Mean(\boldsymbol{D_1}); \boldsymbol{C}_1 = Cov(\boldsymbol{D_1})$$

$$m_1 = Mean(\boldsymbol{Z_1}); v_1 = Var(\boldsymbol{Z_1})$$

The Projected Scalars

$$\boldsymbol{D} = \{\boldsymbol{x}_i : \boldsymbol{x}_i \in \mathbb{R}^d ; i = 1, \dots n\}$$

$$\boldsymbol{Z} = \{z_i : z_i = \boldsymbol{p}^T \boldsymbol{x}_i ; \forall \boldsymbol{x}_i \in \boldsymbol{X} ; z_i \in \mathbb{R}^1\}$$

$$\boldsymbol{p}^T \boldsymbol{p} = 1$$

Evaluating Mean of $\boldsymbol{Z}$

$$\gamma = \frac{1}{n}\sum_{i=1}^{n} z_i \qquad \boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i$$

$$\gamma = f(\boldsymbol{p}, \boldsymbol{\mu})$$

Evaluating Mean of $\boldsymbol{Z}$

$$\gamma = \frac{1}{n}\sum_{i=1}^{n} z_i = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{p}^T \boldsymbol{x_i} = \boldsymbol{p}^T \left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \right)$$

$$\gamma = \boldsymbol{p}^T \boldsymbol{\mu}$$

Evaluating Variance of $\boldsymbol{Z}$

$$\boldsymbol{C} = Cov(\boldsymbol{X}) = \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})^T$$

$$v = Var(Z) = \frac{1}{n}\sum_{i=1}^{n}(z_i - \gamma)^2$$

$$v = g(\boldsymbol{p}, \boldsymbol{C})$$

Evaluating Variance of $\boldsymbol{Z}$

$$v = \frac{1}{n}\sum_{i=1}^{n}(z_i - \gamma)^2 = \frac{1}{n}\sum_{i=1}^{n}(z_i - \gamma)(z_i - \gamma)$$

$$v = \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{p}^T\boldsymbol{x}_i - \boldsymbol{p}^T\boldsymbol{\mu})\left(\boldsymbol{x}_i^T\boldsymbol{p} - \boldsymbol{\mu}^T\boldsymbol{p}\right)$$

Evaluating Variance of $\boldsymbol{Z}$

$$v = \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{p}^T\boldsymbol{x}_i - \boldsymbol{p}^T\boldsymbol{\mu})\left(\boldsymbol{x}_i^T\boldsymbol{p} - \boldsymbol{\mu}^T\boldsymbol{p}\right)$$

$$v = \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{p}^T(\boldsymbol{x}_i - \boldsymbol{\mu})\left(\boldsymbol{x}_i^T - \boldsymbol{\mu}^T\right)\boldsymbol{p}$$

$$v = \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{p}^T(\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})^T\boldsymbol{p}$$

Evaluating Variance of $\boldsymbol{Z}$

$$v = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{p}^T (\boldsymbol{x}_i - \boldsymbol{\mu})\,(\boldsymbol{x}_i - \boldsymbol{\mu})^T \boldsymbol{p}$$

$$v = \boldsymbol{p}^T \left\{ \frac{1}{n}\sum_{i=1}^{n} (\boldsymbol{x}_i - \boldsymbol{\mu})\,(\boldsymbol{x}_i - \boldsymbol{\mu})^T \right\} \boldsymbol{p}$$

$$C = Cov(\boldsymbol{D})$$

$$v = \boldsymbol{p}^T C \boldsymbol{p}$$

Definitions

$$\boldsymbol{D_0} = \{\boldsymbol{x}_i^0 : \boldsymbol{x}_i^0 \in \mathbb{R}^D; i = 1, \dots n_0\}$$

$$\boldsymbol{D_1} = \{\boldsymbol{x}_j^1 : \boldsymbol{x}_j^1 \in \mathbb{R}^D; j = 1, \dots n_1\}$$

$$y(\boldsymbol{x}) = \begin{cases} 1, & \boldsymbol{x} \in \boldsymbol{\omega_1} \\ 0, & \boldsymbol{x} \in \boldsymbol{\omega_0} \end{cases}$$

$$\boldsymbol{Z_0} = \{z_i^0 : z_i^0 = \boldsymbol{p}^T \boldsymbol{x}_i^0; \boldsymbol{x}_i^0 \in \boldsymbol{D_0}\}$$

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$$\boldsymbol{p}^T \boldsymbol{p} = 1$$

$$\boldsymbol{\mu}_0 = Mean(\boldsymbol{D_0}); \boldsymbol{C}_0 = Cov(\boldsymbol{D_0})$$

$$m_0 = Mean(\boldsymbol{Z_0}); v_0 = Var(\boldsymbol{Z_0})$$

$$\boldsymbol{\mu}_1 = Mean(\boldsymbol{D_1}); \boldsymbol{C}_1 = Cov(\boldsymbol{D_1})$$

$$m_1 = Mean(\boldsymbol{Z_1}); v_1 = Var(\boldsymbol{Z_1})$$

Problem Formulation: Step #1

Maximize Distance between Sets $\boldsymbol{Z_0}$ and $\boldsymbol{Z_1}$

$$Maximize: (m_1 - m_0)^2 = \boldsymbol{p}^T \boldsymbol{S_B} \boldsymbol{p}$$

$$(m_1 - m_0)^2 = (\boldsymbol{p}^T \boldsymbol{\mu}_1 - \boldsymbol{p}^T \boldsymbol{\mu}_0)(\boldsymbol{\mu}_1^T \boldsymbol{p} - \boldsymbol{\mu}_0^T \boldsymbol{p})$$

$$= \boldsymbol{p}^T (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{p}$$

Between-Class ($\boldsymbol{D_0}$ and $\boldsymbol{D_1}$) Scatter Matrix

$$\boldsymbol{S_B} = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T$$

Problem Formulation: Step #2

Minimize Net Variances of Sets $\boldsymbol{Z_0}$ and $\boldsymbol{Z_1}$

$$Minimize: v_1 + v_0 = \boldsymbol{p}^T \boldsymbol{S_W} \boldsymbol{p}$$

$$v_1 + v_0 = \boldsymbol{p}^T \boldsymbol{C}_1 \boldsymbol{p} + \boldsymbol{p}^T \boldsymbol{C}_0 \boldsymbol{p} = \boldsymbol{p}^T (\boldsymbol{C}_1 + \boldsymbol{C}_0) \boldsymbol{p}$$

Net Within-Class ($\boldsymbol{D_0}$ and $\boldsymbol{D_1}$) Scatter Matrix

$$\boldsymbol{S_W} = \boldsymbol{C}_1 + \boldsymbol{C}_0$$

LDA: Objective

$$Maximize: \boldsymbol{p}^T \boldsymbol{S_B} \boldsymbol{p} \qquad\qquad Minimize: \boldsymbol{p}^T \boldsymbol{S_W} \boldsymbol{p}$$

$$Maximize: \frac{\boldsymbol{p}^T \boldsymbol{S_B} \boldsymbol{p}}{\boldsymbol{p}^T \boldsymbol{S_W} \boldsymbol{p}} \quad w.r.t. \boldsymbol{p}$$

$$\Downarrow$$

$$g(\boldsymbol{p}; \lambda) = -\boldsymbol{p}^T \boldsymbol{S_B} \boldsymbol{p} + \lambda(\boldsymbol{p}^T \boldsymbol{S_W} \boldsymbol{p} - 1)$$

Recapitulation: KKT Conditions

Minimize: $f(\boldsymbol{x})$ Subject to: $h_i(\boldsymbol{x}) = 0; i = 1, \dots n$

Lagrangian: $\mathrm{g}(\boldsymbol{x}; \alpha_1, \dots \alpha_n) = f(\boldsymbol{x}) + \sum_{i=1}^{n} \alpha_i h_i(\boldsymbol{x})$

KKT Conditions: $\nabla_{\boldsymbol{x}} \mathrm{g}(\boldsymbol{x}; \alpha_1, \dots \alpha_n) = \nabla_{\boldsymbol{x}} f(\boldsymbol{x}) + \sum_{i=1}^{n} \alpha_i \nabla_{\boldsymbol{x}} h_i(\boldsymbol{x}) = 0$

Maximize: $f(\boldsymbol{x})$ Subject to: $h_i(\boldsymbol{x}) = 0; i = 1, \dots n$

Lagrangian: $\mathrm{g}(\boldsymbol{x}; \alpha_1, \dots \alpha_n) = -f(\boldsymbol{x}) + \sum_{i=1}^{n} \alpha_i h_i(\boldsymbol{x})$

KKT Conditions: $\nabla_{\boldsymbol{x}} \mathrm{g}(\boldsymbol{x}; \alpha_1, \dots \alpha_n) = -\nabla_{\boldsymbol{x}} f(\boldsymbol{x}) + \sum_{i=1}^{n} \alpha_i \nabla_{\boldsymbol{x}} h_i(\boldsymbol{x}) = 0$

LDA: Solution

$$g(\boldsymbol{p};\lambda) = -\boldsymbol{p}^T\boldsymbol{S}_B\boldsymbol{p} + \lambda(\boldsymbol{p}^T\boldsymbol{S}_W\boldsymbol{p} - 1)$$

Application of KKT Conditions

$$\nabla_{\boldsymbol{p}} g(\boldsymbol{p};\lambda) = -2\boldsymbol{S}_B\boldsymbol{p} + \lambda(2\boldsymbol{S}_W\boldsymbol{p}) = 0$$

$$\Downarrow$$

$$2\boldsymbol{S}_B\boldsymbol{p} = 2\lambda\boldsymbol{S}_W\boldsymbol{p}$$

$$\Downarrow$$

$$\boldsymbol{S}_W^{-1}\boldsymbol{S}_B\boldsymbol{p} = \lambda\boldsymbol{p}$$

Illustration: $\mathrm{A} = \boldsymbol{u}\boldsymbol{u}^T$ is Rank-1

$$\boldsymbol{u} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \qquad A = \boldsymbol{u}\boldsymbol{u}^T = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -2 \\ -1 & 1 & 2 \\ -2 & 2 & 4 \end{bmatrix}$$

$$eigVec(A) = \begin{bmatrix} -0.41 & 0.71 & -0.58 \\ 0.41 & 0.71 & 0.58 \\ 0.82 & 0 & -0.58 \end{bmatrix}$$

$$eigVal(A) = \begin{bmatrix} 6 & 0 & 0 \end{bmatrix}$$

Illustration: $\mathrm{M} = B * (\boldsymbol{u}\boldsymbol{u}^T)$ is Also Rank-1

$$B = \begin{bmatrix} 3 & 2 & -1 \\ 5 & -7 & 2 \\ 4 & 1 & -3 \end{bmatrix} \longrightarrow eigVal(B) = \begin{bmatrix} 3.51 & -2.48 & -8.03 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -1 & -2 \\ -1 & 1 & 2 \\ -2 & 2 & 4 \end{bmatrix}$$

$$eigVec(M) = \begin{bmatrix} -0.24 & -0.76 & -0.66 \\ -0.64 & 0.35 & -0.75 \\ -0.73 & -0.55 & 0.04 \end{bmatrix}$$

$$M = B * A$$

$$M = \begin{bmatrix} 3 & -3 & -6 \\ 8 & -8 & -16 \\ 9 & -9 & -18 \end{bmatrix} \longrightarrow eigVal(M) = \begin{bmatrix} -23 & 0 & 0 \end{bmatrix}$$

LDA: Solution

$$\boldsymbol{S_W^{-1} S_B p} = \lambda \boldsymbol{p}$$

We are only Interested in the Unit Direction $\boldsymbol{p}$

$$\lambda \boldsymbol{p} = \boldsymbol{S_W^{-1}}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{p} = \boldsymbol{S_W^{-1}}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)\{(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{p}\}$$

$$\left(\frac{\lambda}{\alpha}\right)\boldsymbol{p} = \boldsymbol{S_W^{-1}}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0); \alpha = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{p}$$

Some Scalar α

$$\boldsymbol{p} \propto \boldsymbol{S_W^{-1}}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$$

Example: Input Data

D_1

1	**2**	**3**	**5**	**4**	**6**	**8**
1	**2**	**3**	**4**	**5**	**6**	**7**

D_0

-2	**-1**	**1**	**3**	**4**	**2**	**5**
3	4	5	6	7	8	9

Example: Compute Means

$$\boldsymbol{\mu}_1 = \begin{pmatrix} 4.1429 \\ 4.0000 \end{pmatrix} \qquad \boldsymbol{\mu}_0 = \begin{pmatrix} 1.7143 \\ 6.0000 \end{pmatrix}$$

$$\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0 = \begin{pmatrix} 2.4286 \\ -2.000 \end{pmatrix}$$

Example: Compute Covariances

$$\boldsymbol{C}_1 = \begin{pmatrix} 4.9796 & 4.2857 \\ 4.2857 & 4.0000 \end{pmatrix} \qquad \boldsymbol{C}_0 = \begin{pmatrix} 5.6327 & 4.2857 \\ 4.2857 & 4.0000 \end{pmatrix}$$

$$\boldsymbol{S_W} = \boldsymbol{C}_1 + \boldsymbol{C}_0 = \begin{pmatrix} 10.6122 & 8.5714 \\ 8.5714 & 8.0000 \end{pmatrix}$$

Example: LDA

$$\boldsymbol{u} = \boldsymbol{S}_{\boldsymbol{W}}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0) = \begin{pmatrix} 3.2000 \\ -3.6786 \end{pmatrix}$$

$$\widehat{\boldsymbol{p}} = \frac{\boldsymbol{u}}{\|\boldsymbol{u}\|_2} = \begin{pmatrix} 0.6563 \\ -0.7545 \end{pmatrix}$$

Summary

- LDA: Supervised Algorithm
- Use of Label Information
- Maximize Distance with Compact Groups

Thank You